

MA.912.DP.2.6

Benchmark

MA.912.DP.2.6 Given a scatter plot with a line of fit and residuals, determine the strength and direction of the correlation. Interpret strength and direction within a real-world context.

Benchmark Clarifications:

Clarification 1: Instruction focuses on determining the direction by analyzing the slope and informally determining the strength by analyzing the residuals.

Connecting Benchmarks/Horizontal Alignment

- MA.912.DP.1.1, MA.912.DP.1.3
- MA.912.DP.3.1

Terms from the K-12 Glossary

- Line of Fit
- Numerical Data
- Scatter Plot

Vertical Alignment

Previous Benchmarks

- MA.8.DP.1

Next Benchmarks

- MA.912.DP.2.8
- MA.912.DP.2.9

Purpose and Instructional Strategies

In grade 8, students informally fitted a line to a scatter plot. In Algebra I, students use the slope and residuals of a line of fit to determine the strength and direction of the correlation. In later courses, students will use the slope and residuals to more quantitatively determine the strength of the correlation.

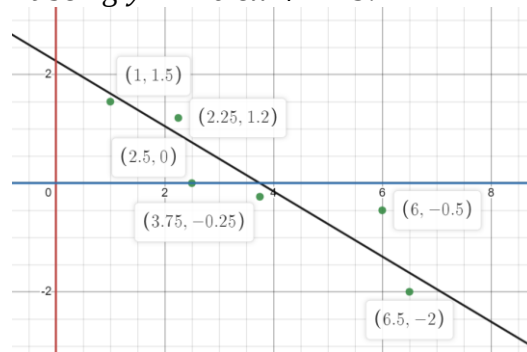
- A residual is a measure of how well a line predicts an individual data point. It can be illustrated by the vertical distance between a data point and the regression line. Each data point has one residual given by the equation $R = D - P$, where R represents the residual, D represents the y -coordinate of data value and P represents the y -coordinate of predicted value. Residuals are positive if data points are above the regression line and negative if data points are below the regression line. If the regression line actually passes through the point, the residual at that point is zero.
 - The slope of the line and the residuals determine the sign and strength of the correlation. A line with a positive slope indicates a positive correlation. A line with a negative slope indicates a negative correlation. Residuals with smaller absolute values indicate stronger correlations. Residuals with larger absolute values indicate weaker correlations.
 - If the slope is close to 0, then the correlation may be considered weak even when the residuals are all small. A slope near zero indicates that the independent variable has little effect on the dependent variable.
- Instruction focuses on real-world contexts and includes the use of technology.
- A residual plot has the residual values on the vertical axis; the horizontal axis displays the x -variable.
- Once residuals are calculated, students can determine the number of positive and negative residuals and the largest and smallest residuals.
- Outliers, which are observed data points that are far from a line of fit, can be determined from the residual as points whose residuals have a large absolute value.

Common Misconceptions or Errors

- Students may not be able to distinguish between a scatter plot and a residual graph.
- Students may forget that residual graphs consist of the ordered pair: independent, residual.
- Students may not be able to determine an appropriate model (linear/nonlinear) from a residual graph.
- Students may think that a correlation is strong when the residuals are small and the slope is close to zero.

Strategies to Support Tiered Instruction

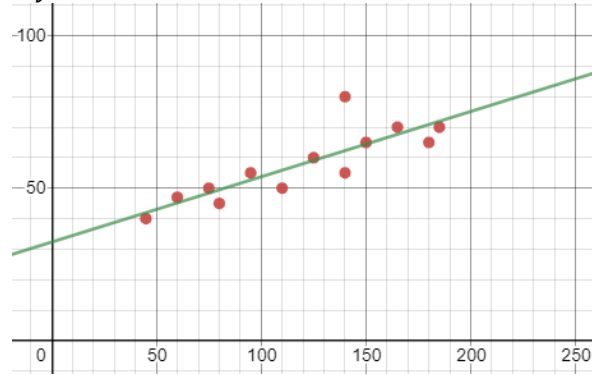
- Teacher co-creates an anchor chart that provides examples of residual graphs for linear models.
- Instruction includes the opportunity to use colors to identify the x - and y -values in the original data and the x - and y -values in the ordered pairs associated with the residual graph, and highlight the x - and y -axis of the residual graph in the same color in order to see the relationship.
 - For example, for the data point $(6, -0.5)$, the residual point is $(6, 0.85)$ based on the line fit being $y = -0.6x + 2.25$.



Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.4.1, MTR.7.1)

Crickets are one of nature's more interesting insects, partly because of their musical ability. In England, the chirping or singing of a cricket was once considered to be a sign of good luck. Crickets will not chirp if the temperature is below 40 degrees Fahrenheit (°F) or above 100 degrees Fahrenheit (°F). A scatter plot is shown with the line of best fit, which can be described by the model $y = 0.214x + 32.317$.



The residuals (r) based on the scatter plot are shown.

Average Number of Chirps (per minute)	Temperature (°F)	Residual
45	40°	-1.95
60	47°	1.84
75	50°	1.62
80	45°	-4.45
95	55°	2.34
110	50°	-5.87
125	60°	0.92
140	55°	-7.29
140	80°	17.7
150	65°	0.57
165	70°	2.35
180	65°	-5.86
185	70°	-1.93

Part A. Determine if the data has a positive or negative correlation.

Part B. Determine the strength of the correlation.

Part C. Compare your answers from Part A and B with a partner.

Part D. Do you notice any possible outliers? Do they effect the judgment of the strength of correlation from Part B?

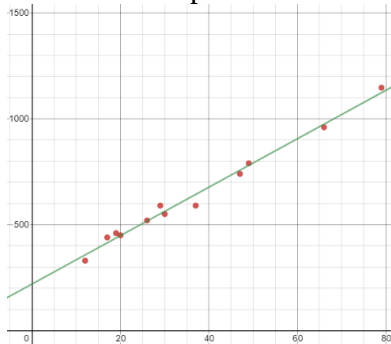
Instructional Items

Instructional Item 1

Based on the data, we know that the line of best fit for the relationship between fat grams and the total calories in fast food using the given data below can be represented by $y = 11.44x + 219.89$. The residuals of this data have been calculated and are represented in the last column.

Sandwich	Total Fat (g)	Total Calories	Residuals
Double Cheeseburger	26	520	2.810
Cheeseburger (1/4 pound)	30	550	-12.93
Cheeseburger (1/3 pound)	47	740	-17.32
Bacon Burger	79	1147	23.78
Bacon Cheeseburger (1/4 pound)	66	960	-14.57
Bacon Cheeseburger (1/3 pound)	49	790	9.814
Hamburger	37	590	-52.97
Fried Chicken Sandwich	29	590	38.51
Grilled Chicken Sandwich	17	440	25.72
Spicy Grilled Chicken Sandwich	19	460	22.85
Roast Beef and Cheese Sandwich	20	450	1.42
Tuna Melt	12	330	-27.11

All of this data is represented on the scatter plot.



Based on this scatter plot and the residuals, interpret the strength and direction of the correlation of the line of fit as it relates to total grams of fat and total calories of fast food.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*